THE GEORGE WASHINGTON UNIVERSITY

WASHINGTON, DC

FINAL PROJECT

Explainable Aspect-Based Sentiment Analysis of Customer Reviews

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DATS 6312: Natural Language Processing for Data Science

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Introduction

Customers frequently leave reviews about products and services they received on online sites. Traditional sentiment analysis assigns a single polarity to an entire review, which can hide important nuance when the writer praises one aspect but criticizes another. Aspect-based sentiment analysis (ABSA) solves this problem by tagging each aspect of the review (e.g., “staff attitude”, “booking ease” or “price value for money”) with its own sentiment. The project described here trains a transformer-based ABSA model on the large FABSA dataset and then uses explainable AI methods to uncover the words driving each sentiment.

Dataset description and preprocessing

FABSA dataset

FABSA is a large, multi-domain aspect-based sentiment analysis dataset created by Jordan Clive and colleagues and hosted on Hugging Face. It contains 10.6 k reviews from three platforms (Trustpilot, Google Play and the iOS App Store) across ten industries. Each review contains a list of (aspect, sentiment) pairs (the aspect category and its sentiment) along with other metadata such as an industry label, an anonymized organisation ID and the full text. FABSA is split into three subsets: train (7.93k reviews), validation (1.06k) and test (1.59 k).

Pre-processing

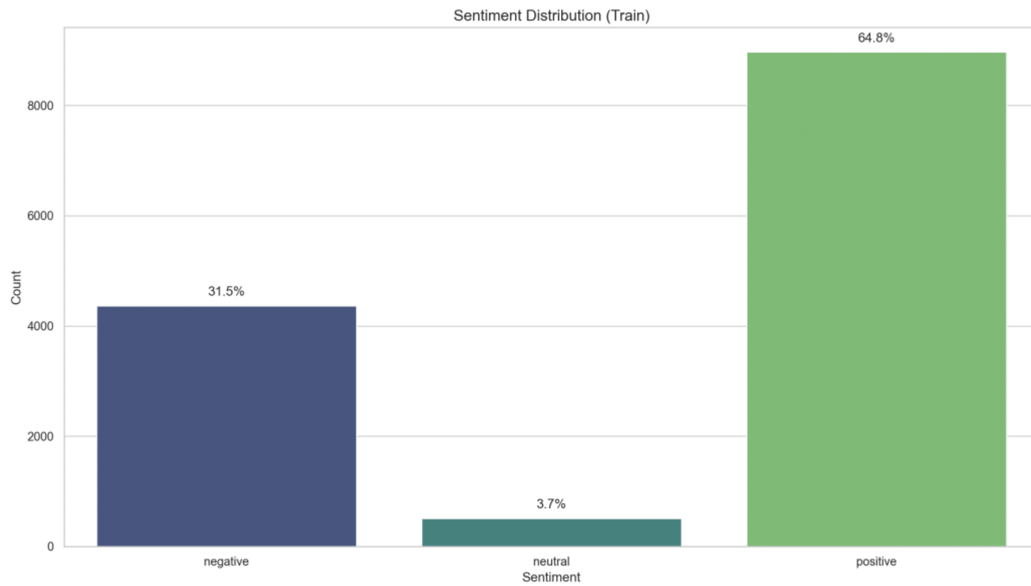
The raw dataset contains reviews with up to 38 aspects per review, anonymized organisation codes (e.g., ORG932) and occasional punctuation artefacts. Following the procedures described in our presentation, we:

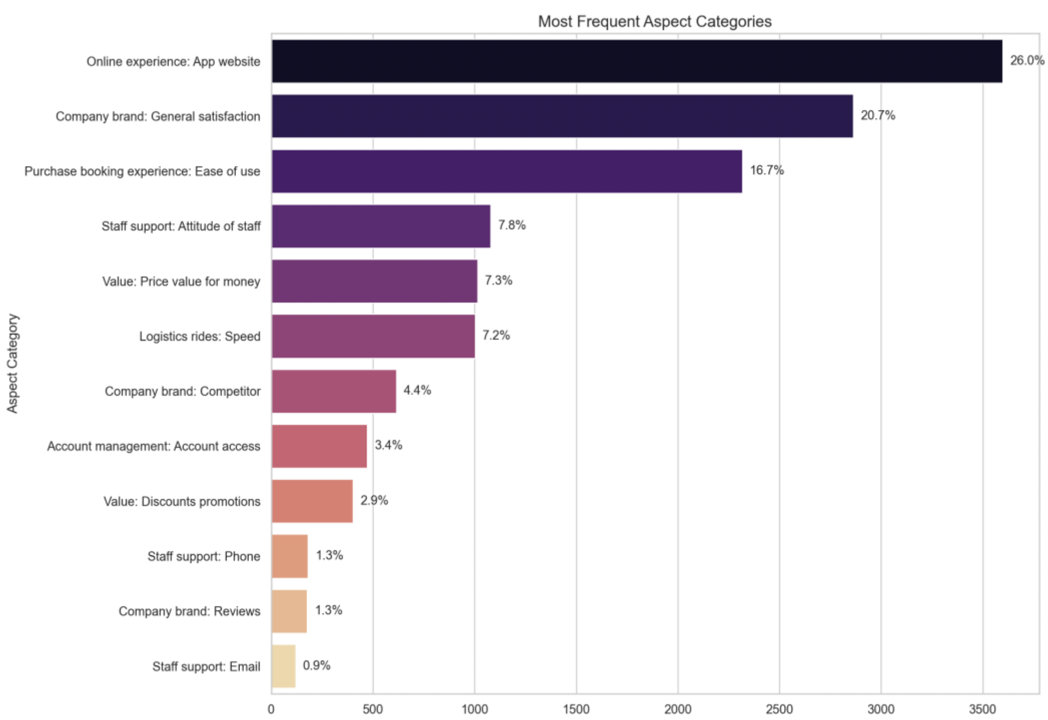
1. Flattened reviews to aspect level: each (text, aspect, sentiment) tuple becomes one training instance. The flattened training split contains 13998 aspect-level rows, the validation split 1858 and the test split 2812.
2. Removed outliers: a small number of reviews with more than eight aspects were excluded to avoid extremely long training sequences.
3. Cleaned text: removed the anonymized org placeholders, converted to lower case and stripped punctuation, removed stopwords and non-letter characters. (stopword removal was applied during the analysis to give stronger insights but not during the model training)

4. Tokenized using the roBERTa BPE tokenizer. Tokenisation is performed at training time by the Hugging Face AutoTokenizer, so no stemming or lemmatisation is applied.

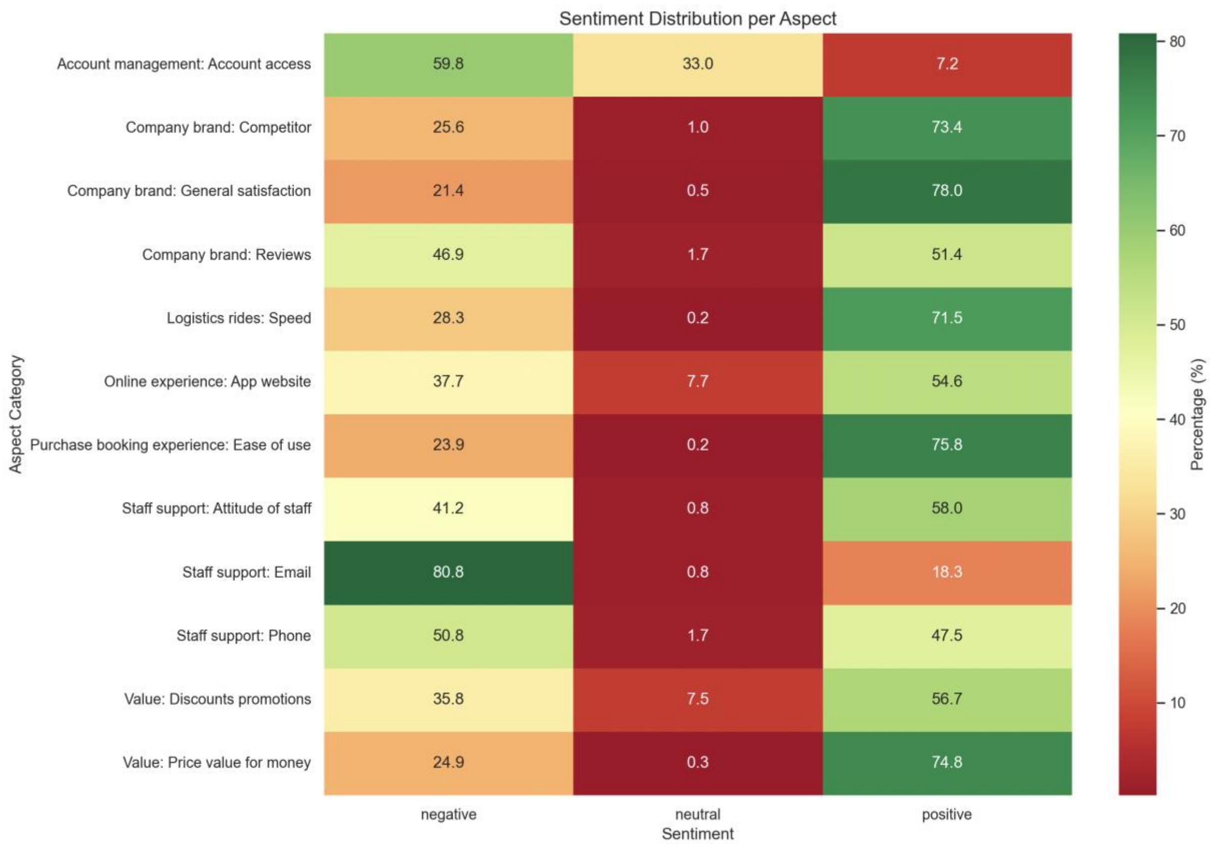
Exploratory analysis

To motivate model design, we inspected the distribution of sentiments and aspects. Sentiment imbalance is pronounced, in the training set approximately 64.8 % of aspect labels are positive, 31.5 % negative and only 3.7 % neutral, as illustrated in the plot below. This imbalance leads to high accuracy but poor recall on the minority neutral class if left unaddressed. On the aspect axis, categories such as "online experience: app website" (26 %), "company brand: general satisfaction" (20.7 %) and "purchase booking experience: ease of use" (16.7 %) dominate the data, whereas others like "staff support: email" appear in fewer than 1 % of samples.





The heatmap below further shows the sentiment distribution for the 12 aspects. For example, "purchase booking experience: ease of use" is overwhelmingly positive (75.8 % positive vs. 23.9 % negative), while "staff support: email" is mostly negative (80.8 % negative vs. 18.3 % positive) and "company brand: competitor" is largely positive (73.4 % positive).



Model and methods

Model architecture

We used the roBERTa-base transformer, a 125-million-parameter encoder pre-trained on 160GB of English text. roBERTa employs byte-pair encoding (BPE) tokenization and deep self-attention layers to capture contextual semantics. For aspect-based sentiment classification the model is fine-tuned to predict one of three sentiment classes (negative, neutral, positive) given both the review text and the aspect category. Input sequences are formatted as *CLS* text *SEP* aspect *SEP*, enabling the model to attend to the aspect category when computing sentiment.

Implementation

We used custom FABSAAspectDataset class wraps the flattened dataset and tokenizes each (text, aspect) pair on the fly. Data loaders build shuffled mini-batches for training and sequential batches for evaluation. The model head is an

AutoModelForSequenceClassification with three output logits. Training uses the AdamW optimizer with a learning rate of 1×10^{-5}, batch size 16, five epochs and maximum sequence length 128 tokens.

Handling class imbalance

Because positive examples dominate the training data, a naive model tends to predict the majority class, yielding high accuracy but poor recall on negative and especially neutral instances. To mitigate this, we computed inverse-frequency class weights $w_c = \frac{N}{n_c}$ (where n_c is the number of samples of class c and N is the total sample count) and pass them to torch.nn.CrossEntropyLoss. In our experiment the computed weights are approximately [3.21 for negative, 27.45 for neutral, 1.53 for positive]. The heavy weight on the neutral class encourages the model to correctly identify the scarce neutral cases.

Results

Model	Class	Precision	Recall	F1-score	Support
Unweighted	Negative	0.90	0.88	0.89	896
	Neutral	0.86	0.79	0.82	91
	Positive	0.94	0.96	0.95	1 825
	Accuracy	-	0.93	-	2 812
	Macro avg	0.90	0.88	0.89	2 812
	Weighted avg	0.93	0.93	0.93	2 812
Class-weighted	Negative	0.84	0.95	0.89	896
	Neutral	0.67	0.85	0.75	91
	Positive	0.98	0.91	0.94	1 825
	Accuracy	-	0.92	-	2 812
	Macro avg	0.83	0.90	0.86	2 812
	Weighted avg	0.93	0.92	0.92	2 812

The class-weighted model achieves slightly lower overall accuracy (0.92 vs. 0.93) and precision but substantially improves recall for the negative class (0.95 vs. 0.88) and the neutral class (0.85 vs. 0.79). In applications where missing negative feedback could be costly, this higher recall is desirable. The weighted model also yields the highest macro-average recall (0.90), demonstrating better balance across classes. Based on these observations we selected the weighted model for further analysis.

Discussion of errors

Despite improvements, neutral reviews remain the most challenging; their support is tiny (91 out of 2 812 test examples) and they often contain information-seeking phrases (“How do I access my card?”) rather than sentiment words. The model sometimes misclassifies short neutrals as positive because roBERTa’s contextual understanding may conflate politeness ("Good afternoon!") with positivity. Future work could augment the neutral class with synthetic examples or use hierarchical loss functions.

Explainability with SHAP

To interpret the weighted model we applied Shapley additive explanations (SHAP) to identify which tokens push the model toward each class. We compute SHAP values on all aspect-level samples and aggregate them by aspect category. Below is a summary of the top tokens (words or subwords) that most strongly contribute to negative, neutral and positive predictions for selected aspects.

Aspect	Top tokens pushing Negative	Top tokens pushing Neutral	Top tokens pushing Positive
Online experience: App website	worst, adapting, disappointing, alerts, wrong, after, laggy, intolerance, even	how, app, disable, afternoon, application, account, good, check	super, great, valuable, quickly, best, excellent, awsome, love, enjoy
Company brand: General satisfaction	difficult, rubbish, really, robbery, fix, would, cant, insurance, sfter, appalling	app, now, how, visa, could, test	awesome, informative, fantastic, amazing, excellent, cool, super, helpful, saving
Purchase/booking experience: Ease of use	down, worst, difficult, let, problems, confused., should, went, wrong	app, anyone, application, platform, hold	fast, easy, ease, easiest, reliable, userfriendly, convenient, simple, great
Staff support: Attitude of staff	slow, issue, incorrect, difficulties, unhelpful, need, not, annoying	account, can, app, address, try, register, looking, am, like, think	helpful, thanks, fast, highly, love, excellent, experienced, valuable, clearly
Value: Price value for money	stealing, insane, frustrating, again, ticket, rubbish, unfortunately, company, tariffs, problems	email, app, accounts, looking, open, confirm	best, awesome, saved, great, excellent, saving, great, useful., love, effective
Logistics rides: Speed	delayed, lagging., hard, slow, withdraw, loaded, short, wish, worst	could, normal, app, details, , enter, may, system	great, beautiful, good, friendly, perfect, liked, love, reliable, efficient, saved

Company brand: Competitor	scandalous., worse, confused., shame, worst, late, only, lack, tells, confusing	how, apps, app, would, apple, page,	best, exceptional., unsurpassed, great, best, highly, brilliant, easy, highly
Account management: Account access	never, missing, shocking, canceled, disabled, reject, worst, rejects	password, id, account, can, account, see, password, hey	accessible, easy, pleased, best, quick, automatically., everywhere, saved, thanks
Value: Discounts promotions	problems, annoying, failed, never, discounts, unable, defective	subscription, could, app, data, download, pls, account, enter	nice, perfect, cool, great, easy, amazing, easier, love
Staff support: Phone	useless., sending, issues, asks, upset, unable, slow, lack, problems	account, sign, app, need, whether, confirm, uninstalled, switch	nice, courteous, efficient, lovely, great, brilliant, excellent, thank
Company brand: Reviews	impossible, bad, still, publicize, without, ruined, annoying, stop, demolish	app, please, greetings, register, downloaded, member, again, select	awesome, fantastic, love, honest, best, happy, good, hesitate, saved, always
Staff support: Email	wrong, problems., missing, scam., bombarded, popped, disappointment, unhelpful	unsubscribe, need, confirm, data, enter, login, details	happy, excellent, saved, solved, welldone, easy, happy, lovely, best

From the table we observe consistent patterns:

- Positive drivers are words like easy, fast, great, excellent, awesome and helpful. Phrases indicating reliability (reliable, efficient) or praise (love, saving) also boost positive predictions. These align with the high proportion of positive labels for aspects such as booking ease and general satisfaction.
- Negative drivers include strongly critical words (worst, stealing, scandalous., worse, useless.) and expressions of frustration (difficult, lagging., slow, annoying) and disappointment. Interestingly, the model picks up morphological cues such as negations as negative signals.
- Neutral drivers often consist of function words and queries like how, app, can, email, subscription, password indicating that neutral reviews tend to be information seeking

rather than evaluative. This explains why the neutral class is rare and easily confused with positive or negative.

Conclusion and future work

This project applied a transformer-based aspect-based sentiment classifier to the FABSA dataset. Our key findings are:

- The dataset exhibits strong sentiment imbalance (65 % positive, 31 % negative, 4 % neutral). Class imbalance significantly affects performance, prompting the use of class-weighted loss.
- A fine-tuned roBERTa-base model achieved 93 % accuracy on the test set. Incorporating inverse-frequency class weights reduced accuracy slightly but increased recall on minority classes (negative and neutral) and yielded the highest macro-average recall. This trade-off is desirable when capturing complaints is more important than maximising overall accuracy.
- SHAP explainability reveals which tokens influence each prediction and highlights aspect-specific drivers of satisfaction or dissatisfaction. Positive sentiment is driven by words like easy, great, helpful and excellent, while negative sentiment is driven by worst, difficult, lagging and scandalous. Neutral reviews often contain information-seeking words, explaining their rarity.
- From a business perspective, the analysis suggests focusing on app reliability, simplifying booking processes, offering competitive pricing and improving customer support. Addressing these areas can convert negative or neutral experiences into positive ones.

Future work may explore:

1. Alternative models: comparing roBERTa with more recent transformer architectures.
2. Data augmentation: generating synthetic neutral examples or oversampling minority classes to further mitigate imbalance.
3. Multi-task learning: jointly training aspect extraction and sentiment classification instead of using pre-extracted aspect labels.
4. Cross-domain transfer: fine-tuning on specific industries and evaluating generalization across domains.

References

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